

# Appendix H - Per-CPU Symbol Index

Entry points, ABI conventions, and reserved memory regions for each CPU. The detailed register descriptions and the per-CPU chapter give the full story; this appendix is the cheat sheet.

## H.1 IE64

| Symbol                 | Value / role                                                                                                                  |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Reset vector           | \$000000 (first instruction at start of RAM).                                                                                 |
| .ie64 image start      | flat image copied at PROG_START = \$001000; execution starts there. Oversized images are refused before memory or PC changes. |
| In-machine assembly    | ASSEMBLE "name" reads the matching assembly source, assembles at PROG_START = \$001000, and writes name.ie64.                 |
| Trap vector base       | \$000400 (8-byte entries, indexed by trap number).                                                                            |
| Supervisor stack       | grows down from \$0A0000.                                                                                                     |
| User stack (R31)       | grows down from BASIC's per-program stack region.                                                                             |
| Call ABI               | Arguments R8-R15; result R8; caller-saved R1-R7; callee-saved R16-R30; R0 = 0; R31 = SP.                                      |
| FPU regs               | F0-F15; FP32 values, with double operations using register pairs. F0-F7 are argument / result registers by convention.        |
| BASIC text / variables | Dynamic IE64 BASIC arena, discovered through BASIC state pointers.                                                            |
| BASIC stack            | Dynamic IE64 BASIC reservation near the top of the low32 resident window, capped below \$10000000.                            |

## H.2 IE32

| Symbol       | Value / role                                                                                     |
|--------------|--------------------------------------------------------------------------------------------------|
| Reset vector | \$000000.                                                                                        |
| Stack base   | \$09F000 (STACK_START); grows down.                                                              |
| Timing       | WAIT n for short microsecond delays; use device status or interrupts for frame and audio timing. |
| Call ABI     | Arguments A,X,Y,Z; result A; B-W caller-saved; stack via PUSH / POP.                             |

## H.3 6502

| Symbol           | Value / role                  |
|------------------|-------------------------------|
| Reset vector     | \$FFFC (low) / \$FFFD (high). |
| IRQ / BRK vector | \$FFFE / \$FFFF.              |
| NMI vector       | \$FFFA / \$FFFB.              |

| Symbol          | Value / role                                                                  |
|-----------------|-------------------------------------------------------------------------------|
| Stack page      | \$0100-\$01FF, indexed by S.                                                  |
| Zero page       | \$0000-\$00FF.                                                                |
| Bank registers  | \$F700-\$F705, \$F7F0.                                                        |
| MMIO aperture   | \$F000-\$FFF9, mirrors \$F0000-\$F0FF9.                                       |
| VGA C64-style   | \$D700-\$D70D.                                                                |
| ULA paged port  | \$D800-\$D817.                                                                |
| PSG / SID       | \$D400-\$D40F, \$D500-\$D55F.                                                 |
| POKEY           | \$D200-\$D20A.                                                                |
| TED audio       | \$D600-\$D605.                                                                |
| MIDI/MUS player | \$FBA0-\$FBBF.                                                                |
| Live MIDI port  | \$FBF4-\$FBF6; data and control writes are port writes, not RAM shadow bytes. |

## H.4 Z80

| Symbol           | Value / role                                          |
|------------------|-------------------------------------------------------|
| Reset vector     | \$0000.                                               |
| NMI vector       | \$0066.                                               |
| RST n            | n * 8 (\$00, \$08, ..., \$38).                        |
| IM 2 vector base | (I << 8)   (data byte from device).                   |
| Bank registers   | port-mapped through bus translation at \$F700-\$F705. |
| MMIO aperture    | \$F000-\$FFF9.                                        |
| PSG / AY ports   | \$F0 select, \$F1 data.                               |
| TED audio ports  | \$F2, \$F3.                                           |
| POKEY ports      | \$D0, \$D1.                                           |
| SID ports        | \$E0, \$E1.                                           |
| SN76489 ports    | \$E4 data, \$E5 status.                               |
| VGA ports        | \$A0-\$AD.                                            |

## H.5 M68K (MC68020-Class)

| Symbol        | Value / role                                          |
|---------------|-------------------------------------------------------|
| Reset vector  | \$0000.0000 (initial SSP) / \$0000.0004 (initial PC). |
| Vector table  | \$0000.0000-\$0000.03FC (256 entries, 4 bytes each).  |
| Bus error     | vector 2.                                             |
| Address error | vector 3.                                             |

| Symbol              | Value / role                                                                                                                                                                                 |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FPU data registers  | FP0-FP7, 68881-style extended memory format for FMOVEM.                                                                                                                                      |
| FPU control regs    | FPCR, FPSR, FPIAR, moved as longwords.                                                                                                                                                       |
| FPU state frame     | FSAVE idle frame version \$1F, payload size \$18; FRESTORE consumes the frame.                                                                                                               |
| Illegal             | vector 4.                                                                                                                                                                                    |
| Zero divide         | vector 5.                                                                                                                                                                                    |
| CHK                 | vector 6.                                                                                                                                                                                    |
| Trapv               | vector 7.                                                                                                                                                                                    |
| Privilege violation | vector 8.                                                                                                                                                                                    |
| Trace               | vector 9 (trace bits are stored; this chip does not raise trace traps).                                                                                                                      |
| Line A              | vector 10.                                                                                                                                                                                   |
| Line F              | vector 11.                                                                                                                                                                                   |
| TRAP #n             | vectors 32-47.                                                                                                                                                                               |
| Auto-vector IRQs    | vectors 25-31.                                                                                                                                                                               |
| Call ABI            | Arguments on stack; D0 / A0 caller-saved; D2-D7 / A2-A6 callee-saved.                                                                                                                        |
| Flat image loading  | Raw image bytes are deposited at the M68K entry point. The loader leaves MMIO apertures and the stack guard hole untouched; normal programme stores still reach MMIO after execution starts. |

## H.6 x86 (8086 + 386 extensions, real-mode only)

| Symbol           | Value / role                                                                                                                                                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reset vector     | EIP = 0, CS = 0, DS = ES = SS = 0 (flat, not the 8086 F000:F000 boot vector).                                                                                |
| 8086 image start | bytes loaded at \$00000000, execution starts with EIP = 0; use an entry stub at 0 if the body lives elsewhere.                                               |
| IVT              | \$0000-\$03FF (256 entries, 4 bytes each: offset + segment).                                                                                                 |
| MMIO data access | \$000F0000-\$000FFFFF is native MMIO. Data accesses in \$F000-\$FFFF mirror \$000F0000-\$000F0FFF. Instruction fetch at \$F000 reads flat RAM at \$0000F000. |
| Stack            | SS:ESP, segments are zero so the stack lives in flat RAM.                                                                                                    |
| Call ABI         | Caller pushes arguments right-to-left; EAX returns; EBX, ESI, EDI, EBP callee-saved; ECX, EDX caller-saved.                                                  |
| BIOS-style ints  | reserved; no BIOS ROM is provided. The IVT is initialised to a default IRET routine.                                                                         |

Real-mode 20-bit physical address calculation ( $\text{seg} \ll 4$ ) + ofs is part of the CPU address path. The 32-bit linear form (the result of the calculation) is what reaches the bus. Programs that use 32-bit immediate addressing reach the full flat address space directly.

## H.7 Cross-CPU bus addresses (shared)

These addresses are the same in every CPU's 32-bit view of the bus. The 8-bit CPUs reach them through the bank-window mechanism described in Chapters 27 and 28.

| Address                 | Meaning                                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|
| \$F0700                 | TERM_OUT.                                                                                                               |
| \$2700-\$27FF           | 6502/Z80 terminal, input, mouse, and RTC aliases after selecting TERM_IO_BANK into bank 1.                              |
| TERM_IO_BANK            | Bank-1 value for 6502/Z80 access to the terminal/input/RTC block.                                                       |
| SET_TERMINAL_BANK       | 6502/Z80 include-file helper that selects TERM_IO_BANK.                                                                 |
| \$F075C/\$F0760         | RTC_MONO_USEC_LO / RTC_MONO_USEC_HI, monotonic elapsed microseconds.                                                    |
| \$F0BA0                 | MIDI/MUS file-player block.                                                                                             |
| \$F0BF4                 | IE_MIDI_LIVE_DATA, byte-wide live MIDI stream input; writes do not shadow into RAM.                                     |
| \$F0BF5                 | IE_MIDI_LIVE_STATUS, bit 0 set means live MIDI active.                                                                  |
| \$F0BF6                 | IE_MIDI_LIVE_CTRL, bit 0 reset; writes do not shadow into RAM.                                                          |
| \$F1400                 | HOST appliance block.                                                                                                   |
| \$F2200                 | File I/O block. FILE_READ_MAX is at \$F221C; IE64 FILE_DATA_PTR64 is at \$F22B0.                                        |
| \$F2300                 | Media loader.                                                                                                           |
| \$F2320                 | RUN loader block.                                                                                                       |
| \$F2340                 | Coprocessor.                                                                                                            |
| \$F238C                 | COPROC_INSTANCE, worker instance selector.                                                                              |
| \$F23B0                 | Coprocessor selected-worker statistics.                                                                                 |
| \$F25A0                 | Coprocessor capability, version, window, ring, and instance-state discovery.                                            |
| \$F2400                 | SysInfo.                                                                                                                |
| \$F2410                 | SYSINFO_FEATURES; bit 3, SYSINFO_FEATURE_V00D00_TEX_SLOTS, reports retained Voodoo texture slots.                       |
| \$F2414/\$F2418         | SYSINFO_LOW_WINDOW_LO / SYSINFO_LOW_WINDOW_HI, dense low-RAM window size.                                               |
| \$F8000                 | Voodoo 3D.                                                                                                              |
| \$F833C/\$F8340/\$F8344 | V00D00_CMD_PTR / V00D00_CMD_COUNT / V00D00_CMD_SUBMIT, Voodoo command-stream replay.                                    |
| \$000000001/\$000000002 | V00D00_CMD_SUBMIT_REPLAY / V00D00_CMD_SUBMIT_REPLAY_LE, big-endian or little-endian command-stream pairs.               |
| \$F8350/\$F8354         | V00D00_TEX_SLOT / V00D00_TEX_BIND, retain an upload by identifier and bind it again. V00D00_TEX_SLOT_NONE is \$FFFFFFF. |

The coprocessor mailbox occupies \$790000 to \$792FFF, with twelve \$400-byte ring slots. M68K receives its selected ring base in A4, x86 in EBP, and IE64 in R30 at worker entry. M68K, x86, and IE64 support instances 0 and 1; IE32, 6502, and Z80 support instance 0.